

InDrive Analytics

Operational Performance Analysis for a Ride-Hailing Platform

Tools Used: Excel • SQL • Power BI

Domain: Transportation | Mobility | Operations Analytics

Project Level: End-to-End Business Intelligence Project

Disclaimer: This project is created solely for learning and portfolio purposes. The dataset used in this project is sourced from a publicly available YouTube tutorial and is **not an official dataset of inDrive**. The company name is used only to simulate a real-world business analytics scenario.

All analysis and dashboards presented in this project are created by the author.

Dataset link : (<https://topvarsity.in/ola-data-analyst-project/>).

1. Business Background

InDrive is a rapidly growing ride-hailing platform connecting customers with drivers across multiple cities. While booking volumes continue to increase, the Operations team has observed a decline in overall service quality due to increasing ride cancellations, incomplete trips, longer waiting times, inconsistent driver performance, and revenue leakage.

Leadership wants a centralized analytics solution that provides complete visibility into booking operations, customer behavior, driver performance, operational efficiency, and revenue trends to support strategic decision-making.

2. Problem Statement

Indrive currently lacks a centralized reporting system to monitor operational performance and identify the factors affecting ride completion, customer satisfaction, and revenue generation.

Business leaders need a comprehensive analytics solution to answer critical operational questions such as:

- Why are bookings getting cancelled?
- Which areas experience the highest operational delays?
- Which vehicle categories perform best?
- Where is revenue being lost?

- Which customers and drivers require attention?
- Which operational improvements can increase ride completion rates?

The goal is to transform raw booking data into actionable business insights that improve operational efficiency and customer experience.

3. Business Objectives

The project aims to:

- Increase Ride Completion Rate
- Reduce Customer & Driver Cancellations
- Identify Operational Bottlenecks
- Improve Driver Performance Monitoring
- Enhance Customer Experience
- Optimize Fleet Utilization
- Minimize Revenue Leakage
- Support Management with Interactive Business Dashboards

Column	Description
Date	Booking Date
Time	Booking Time
Booking ID	Unique Booking Identifier
Booking Status	Success / Cancelled / Incomplete
Customer ID	Customer Identifier
Vehicle Type	Mini, Sedan, SUV, Bike, etc.
Pickup Location	Ride Origin
Drop Location	Ride Destination
V_TAT	Vehicle Turnaround Time

Column	Description
C_TAT	Customer Waiting Time
Cancelled Rides by Customer	Customer Cancellation Indicator
Cancelled Rides by Driver	Driver Cancellation Indicator
Incomplete Rides	Incomplete Ride Indicator
Incomplete Ride Reason	Reason for Incomplete Ride
Booking Value	Ride Fare
Payment Method	Cash / Card / UPI / Wallet
Ride Distance	Ride Distance (km)
Driver Rating	Driver Rating
Customer Rating	Customer Rating

5. Key Performance Indicators (KPIs)

Booking KPIs

- Total Bookings
- Successful Bookings
- Cancelled Bookings
- Incomplete Rides
- Ride Completion Rate
- Cancellation Rate
- Incomplete Ride Rate

Revenue KPIs

- Total Revenue
- Average Booking Value
- Revenue Lost Due to Customer Cancellations

- Revenue Lost Due to Driver Cancellations
 - Revenue Lost Due to Incomplete Rides
-

Customer KPIs

- Average Customer Rating
 - Average Customer Waiting Time (C_TAT)
 - Customer Cancellation Rate
 - Repeat Customer Bookings
-

Driver KPIs

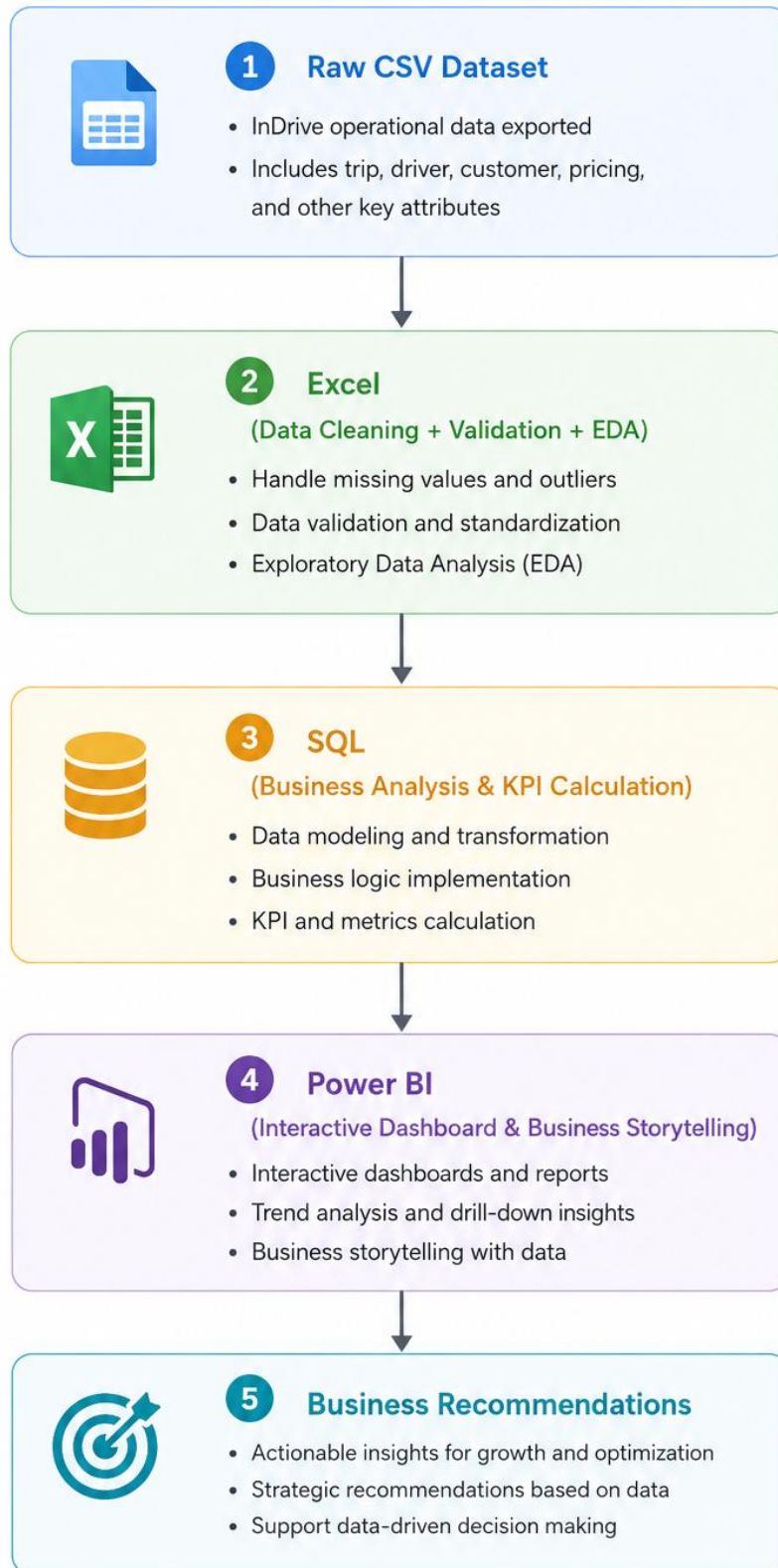
- Average Driver Rating
 - Driver Cancellation Rate
 - Driver Performance Score
 - Average Ride Distance Covered
-

Operational KPIs

- Average Vehicle Turnaround Time (V_TAT)
 - Average Customer Turnaround Time (C_TAT)
 - Peak Booking Hour
 - Peak Booking Day
 - Most Active Pickup Locations
 - Most Active Drop Locations
-

6. Project Workflow

End-to-end workflow for InDrive Data Analytics Project



7. Excel Phase

Objective

Prepare high-quality data for analysis and identify initial trends before moving into SQL.

Data Cleaning

- Remove duplicate bookings
- Handle missing values
- Standardize booking status
- Standardize vehicle type names
- Standardize payment methods
- Convert Date format
- Convert Time format
- Validate ratings
- Detect outliers
- Verify booking IDs
- Check invalid ride distances
- Validate booking values

Exploratory Data Analysis (EDA)

Booking Analysis

- Total Bookings by Date
- Booking Trend
- Booking Status Distribution

Vehicle Analysis

- Vehicle Usage
- Vehicle Booking Share

Revenue Analysis

- Revenue Trend by booking value
- Average Booking Value

Customer Analysis

- Customer Ratings Distribution
- Waiting Time Distribution

Operational Analysis

- Ride Distance Distribution
- Pickup Distribution
- Drop Distribution

EXCEL PIVOTS :

Row Labels	Count of Booking_ID	VEHICLE BOOKING SHARE						Row Labels	Sum of Booking_Value
01-Jul	3351							01-Jul	₹ 18,59,732.0
02-Jul	3388							02-Jul	₹ 18,83,051.0
03-Jul	3358							03-Jul	₹ 18,50,847.0
04-Jul	3274							04-Jul	₹ 18,31,795.0
05-Jul	3306							05-Jul	₹ 17,63,798.0
06-Jul	3281							06-Jul	₹ 18,39,535.0
07-Jul	3223							07-Jul	₹ 17,53,264.0
08-Jul	3313							08-Jul	₹ 18,03,985.0
09-Jul	3358							09-Jul	₹ 18,59,517.0
10-Jul	3375							10-Jul	₹ 18,38,171.0
11-Jul	3242							11-Jul	₹ 17,78,386.0
12-Jul	3382							12-Jul	₹ 18,22,687.0
13-Jul	3409							13-Jul	₹ 18,23,095.0
14-Jul	3307							14-Jul	₹ 17,64,153.0
15-Jul	3318							15-Jul	₹ 18,73,200.0
16-Jul	3371							16-Jul	₹ 18,39,892.0
17-Jul	3343							17-Jul	₹ 18,53,330.0
18-Jul	3359							18-Jul	₹ 18,68,848.0
19-Jul	3319							19-Jul	₹ 17,48,603.0
20-Jul	3289							20-Jul	₹ 18,01,315.0
21-Jul	3237							21-Jul	₹ 17,52,365.0
22-Jul	3377							22-Jul	₹ 18,12,812.0
23-Jul	3377							23-Jul	₹ 18,32,123.0
24-Jul	3286							24-Jul	₹ 17,89,575.0
25-Jul	3361							25-Jul	₹ 18,75,293.0
26-Jul	3338							26-Jul	₹ 18,66,231.0
27-Jul	3362							27-Jul	₹ 18,37,516.0
28-Jul	3246							28-Jul	₹ 17,82,082.0
29-Jul	3370							29-Jul	₹ 19,24,027.0
30-Jul	3432							30-Jul	₹ 18,96,548.0
31-Jul	3072							31-Jul	₹ 17,08,838.0
Grand Total	103024							Grand Total	₹ 5,65,34,614.0

Count of Booking_Status	TOP 10 PICK-UP LOCATIONS & DROP LOCATIONS					
Canceled by Customer	10.19%					
Canceled by Driver	17.89%					
Driver Not Found	9.83%					
Success	62.09%					
Grand Total	100.00%					

Count of Booking_ID	Count of Booking_ID	Count of Booking_ID	Count of Booking_ID
14.44% Cox Town	2100 Hebbal	2097	
14.38% Tumkur Road	2105 Vijayanagar	2103	
14.32% Vijayanagar	2113 Marathahalli	2104	
14.28% Ramamurthy Nagar	2116 Koramangala	2105	
14.23% Basavanagudi	2120 Sarjapur Road	2108	
14.22% Sahakar Nagar	2126 HSR Layout	2117	
14.12% Indiranagar	2133 Hennur	2120	
RT Nagar	2135 MG Road	2128	
Yeshwanthpur	2139 Mysore Road	2148	
Banashankari	2201 Peenya	2159	
Grand Total	21288 Grand Total	21189	

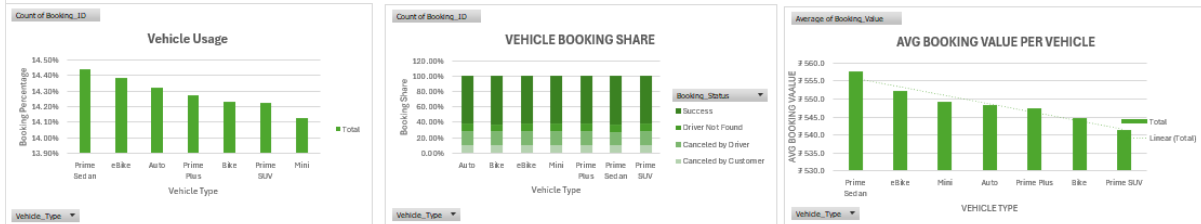
Count of C_TAT	Average of Booking_Value	Count of Customer_Rating
2478	₹ 557.8 3	1651
2560	₹ 552.2 3.1	3130
2626	₹ 549.1 3.2	3222
2627	₹ 548.4 3.3	3194
2525	₹ 547.4 3.4	3210
2519	₹ 544.7 3.5	3275
2642	₹ 541.3 3.6	3168
2664	₹ 548.8 3.7	3264
2541	3.8	3102
2528	3.9	3286
2542	45498 4	3259
2554	8060 4.1	3172
2480	7990 4.2	3173
2556	5654 4.3	3152
2631	5540 4.4	3210
2576	5618 4.5	3173
2585	5677 4.6	3129
2588	5551 4.7	3126
2577	5567 4.8	3216
2565	7869 4.9	3289
2431	103024 5	1566
2537	null	39057
2513	Grand Total	103024
2548		
2574		
39057		
Grand Total		

EXCEL DASHBOARD :

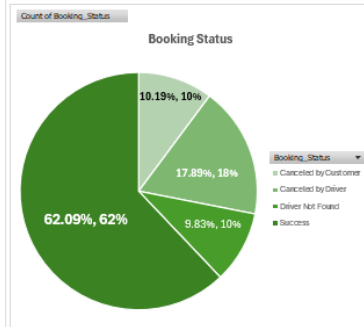
InDrive Bookings Performance Dashboard - July



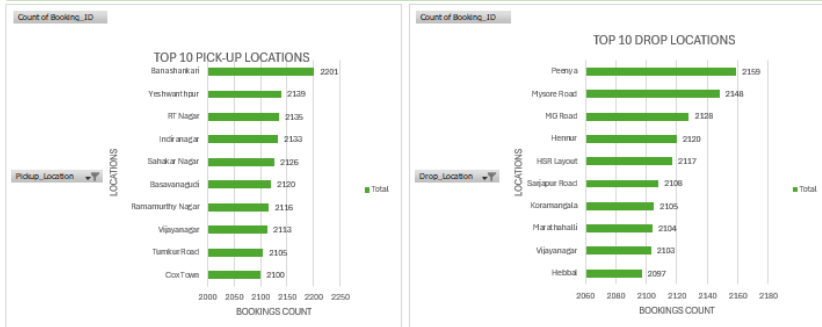
FLEET USAGE & BOOKING SHARE



BEHAVIORAL INSIGHTS



OPERATIONAL ANALYSIS



CUSTOMER ANALYSIS BY RATINGS & WAITING TIME



Note: The dataset does not include a dedicated Customer Waiting Time field. Therefore, C_TAT was used as the closest available customer time-related metric for this analysis.

8. SQL Phase

The SQL analysis is divided into **5 business modules**, moving from descriptive reporting to advanced analytical questions.

Module 1 — Booking Performance

Business Questions

1. What is the total number of bookings?

A. Query :

```
select count(booking_id) as total_bookings from bookings;
```

Result :

	total_bookings
▶	103024

Business Insight :

The platform recorded **103,024 total bookings** during the analysis period, indicating the overall demand for ride-hailing services.

2. How many rides were successfully completed during the analysis period?

A. Query :

```
select count(booking_id) as successful_bookings  
from bookings where booking_status='Success';
```

Result :

	successful_bookings
▶	63967

Business Insight :

Out of 103,024 total bookings, 63,967 rides were successfully completed, resulting in a **62.09% ride completion rate**. While a majority of bookings were fulfilled successfully, improving cancellation rates could further increase completed rides and overall revenue

3. How many ride bookings were cancelled during the analysis period?

A. Query :

```
select count(booking_id) as canceled_bookings from bookings
```

where booking_status in ('Canceled by driver','Canceled by customer', 'Driver not found');

Result :

	canceled_bookings
▶	39057

Business Insight :

A total of **39,057 bookings were cancelled**, representing **37.91% of all bookings**. This indicates a considerable number of rides did not reach completion. Reducing customer and driver cancellations through better driver availability, improved ride matching, and operational efficiency can increase ride completion, improve customer satisfaction, and drive higher revenue.

4. How many rides remained incomplete during the analysis period?

A. Query :

```
SELECT  
  
    COUNT(CASE WHEN incomplete_rides = 'yes' THEN 1 END) AS yes_count,  
  
    COUNT(CASE WHEN incomplete_rides IS NULL THEN 1 END) AS null_count  
  
FROM bookings;
```

Result :

	yes_count	null_count
▶	3926	39057

Business Insight :

Out of all bookings, **3,926 rides were marked as incomplete**, indicating that a portion of rides could not be successfully completed after booking. These incomplete rides may result from issues such as driver no-shows, customer unavailability, or operational disruptions. Investigating the root causes and implementing corrective measures can improve ride completion rates, enhance customer satisfaction, and reduce revenue leakage.

5. What is the overall ride completion rate, and does it indicate operational efficiency?

A. Query :

```
select round(  
  
    count(  

```

```
case when booking_status='Success' then 1 end)*100.0/count(*), 2) as  
ride_completion_rate from bookings;
```

Result :

	ride_completion_rate
▶	62.09

Business Insight :

The analysis shows an overall ride **completion rate of 62.09%**, meaning approximately **6 out of every 10 bookings are successfully completed**. While a majority of bookings are fulfilled, nearly **38% of bookings do not reach successful completion due to cancellations or incomplete rides**. This represents a significant opportunity to improve operational efficiency, increase customer satisfaction, and maximize revenue by reducing ride failures.

6. What is the overall cancellation rate, and what does it reveal about the reliability of the booking process?

A. Query :

```
select round(  
    count(  
        case when booking_status in ('Canceled by driver','Canceled by customer',  
        'Driver not found') then 1 end)*100.0/count(*), 2)  
as cancellation_rate from bookings;
```

Result :

	cancellation_rate
▶	37.91

Business Insight :

The analysis indicates an **overall cancellation rate of 37.91%, meaning nearly 4 out of every 10 bookings** are cancelled before completion. This is a significant operational concern, as a high cancellation rate directly impacts revenue, customer satisfaction, and driver utilization. Reducing cancellations through improved driver availability, better ride matching, and proactive customer communication can increase ride completion rates and improve overall business performance.

7. What percentage of total bookings resulted in incomplete rides, and does this indicate an operational concern?

A. Query :

```
select round(count(case when incomplete_rides is null then 1 end)*100.0/count(*), 2) as incomplete_ride_rate from bookings;
```

Result :

	incomplete_ride_rate
▶	37.91

Business Insight :

37.91% of bookings were incomplete, indicating that more than one in three ride requests did not reach successful completion. This suggests opportunities to improve ride fulfillment by addressing factors such as driver availability, customer cancellations, route allocation, and operational efficiency. Reducing the incomplete ride rate can directly improve customer satisfaction, ride completion, and overall revenue.

8. What is the average booking value per ride, and how effectively is each completed booking contributing to overall revenue?

A. Query :

```
select avg(booking_value) as avg_booking_value from bookings;
```

Result :

	avg_booking_value
▶	548.7519

Business Insight :

The average booking value is **₹548.75 per ride**, indicating the typical revenue generated from each booking. Increasing this metric through premium ride categories, dynamic pricing during peak demand, route optimization, or value-added services can improve overall revenue without necessarily increasing the number of bookings.

9. What is the average ride distance per booking, and what does it indicate about customer travel patterns?

A. Query :

```
select avg(ride_distance) as avg_ride_distance from bookings;
```

Result :

	avg Ride distance
▶	14.1899

Business Insight :

The average ride distance is **14.18 km per booking**, indicating that customers primarily use the platform for medium-distance travel. This insight can help optimize driver allocation, estimate fuel and operational costs, and support pricing strategies to improve service efficiency and profitability.

10. How do booking volumes vary across different days of the week, and which days experience the highest demand?

A. Query :

```
select date(booking_date) as day, count(booking_id) as total_bookings
from bookings group by date(booking_date) order by date(booking_date);
```

Result :

	day	total_bookings
▶	2024-07-01	3351
	2024-07-02	3388
	2024-07-03	3358
	2024-07-04	3274
	2024-07-05	3306
	2024-07-06	3281
	2024-07-07	3223
	2024-07-08	3313
	2024-07-09	3359

Business Insight :

The booking trend by day reveals clear variations in customer demand throughout the week. Identifying peak booking days enables the business to optimize driver allocation, improve fleet availability, and implement demand-based pricing strategies. Aligning operational resources with daily demand patterns can reduce waiting times, improve ride completion rates, and enhance customer satisfaction.

11. How does booking demand vary across different hours of the day, and which hours experience the highest booking activity?

A. Query :

```
select hour(time) as booking_hour, count(booking_id) as total_bookings
from bookings group by hour(time) order by hour(time);
```

Result :

	booking_hour	total_bookings
▶	0	4318
	1	4329
	2	4305
	3	4201
	4	4263
	5	4264
	6	4271
	7	4304

Result 11 ✕

Business Insight :

The booking trend by hour highlights the periods of highest customer demand throughout the day. Identifying peak booking hours enables the business to optimize driver deployment, reduce customer waiting times, and improve ride completion rates. These insights also support dynamic pricing strategies and more efficient resource allocation during high-demand periods.

12. Which vehicle types receive the highest number of bookings, and how is customer demand distributed across the available vehicle categories?

A. Query :

```
select vehicle_type, count(booking_id) as total_bookings from bookings
group by vehicle_type order by total_bookings desc;
```

Result :

	vehicle_type	total_bookings
▶	Prime Sedan	14877
	eBike	14816
	Auto	14755
	Prime Plus	14707
	Bike	14662
	Prime SUV	14655
	Mini	14552

Business Insight :

Prime Sedan recorded the highest number of bookings (14,877), followed closely by eBike (14,816) and Auto (14,755). The booking distribution across all vehicle types is relatively balanced, indicating consistent customer demand across the fleet. This suggests that maintaining adequate driver availability for all vehicle categories while giving slight priority to higher-demand segments can improve ride fulfillment and customer satisfaction.

Module 2 — Customer Analytics

Business Questions

1. Who are the top 10 customers based on the number of bookings, and how can they be leveraged to improve customer retention and revenue?

A. Query :

```
select customer_id, count(booking_id) as total_bookings from bookings  
  
group by customer_id  
  
order by total_bookings desc, customer_id asc limit 10;
```

Result :

	customer_id	total_bookings
▶	CID954071	5
	CID189965	4
	CID199017	4
	CID201733	4
	CID263459	4
	CID266327	4
	CID268274	4
	CID288207	4
	CID309168	4
	CID329193	4

Business Insight :

The top 10 customers have completed the highest number of bookings, indicating strong engagement and loyalty to the platform. These high-value customers present an opportunity for targeted retention strategies such as loyalty rewards, personalized offers, and premium membership benefits. Retaining these customers can increase repeat bookings, strengthen customer lifetime value, and drive long-term revenue growth.

2. Which customers contribute the highest booking value, and how can the business retain these high-value customers to maximize revenue?

A. Query :

```
select customer_id, sum(booking_value) as total_customer_booking_value  
  
from bookings group by customer_id  
  
order by total_customer_booking_value desc limit 10;
```

Result :

	customer_id	total_customer_booking_value
▶	CID785112	8025
	CID308763	6281
	CID734557	6177
	CID353074	6110
	CID836942	6019
	CID868113	5986
	CID389515	5970
	CID695232	5962
	CID497169	5949
	CID152235	5948

Business Insight :

Customer CID785112 generated the highest booking value (₹8,025), followed by CID308763 (₹6,281) and other high-value customers. These customers contribute significantly to overall revenue and should be prioritized through loyalty programs, personalized offers, premium benefits, and targeted retention campaigns. Retaining high-value customers can increase customer lifetime value, encourage repeat bookings, and drive sustainable revenue growth.

3. Which customers have the highest number of ride cancellations, and what actions can be taken to reduce repeated cancellations?

A. Query :

```
select customer_id, sum(case when booking_status in ('Canceled by driver','Canceled by customer', 'Driver not found') then 1 end) as
highest_cancellations

from bookings group by customer_id

order by highest_cancellations desc;
```

Result :

	customer_id	highest_cancellations
▶	CID340854	4
	CID381415	4
	CID460299	4
	CID834203	3
	CID253884	3
	CID398986	3

Business Insight :

The analysis identifies customers with the highest cancellation counts, indicating a higher likelihood of abandoned or unsuccessful bookings. These customers should be further analyzed to determine the root causes of cancellations, such as long waiting times, pricing concerns, or driver unavailability. Targeted retention

strategies, personalized offers, and improved service quality can help reduce repeat cancellations, improve customer satisfaction, and increase ride completion rates.

4. Which customers experienced the longest waiting time before their ride was assigned or started?

A. Query :

```
select customer_id, max(c_tat) as longest_waiting_time from bookings where c_tat is not null group by customer_id order by longest_waiting_time desc limit 10;
```

Result :

	customer_id	longest_waiting_time
▶	CID740890	145
	CID971865	145
	CID184275	145
	CID706421	145
	CID756720	145
	CID953467	145

TIME MENTIONED IN SECONDS.

Business Insight :

The analysis shows that the maximum customer waiting time was approximately **145 seconds (2 minutes 25 seconds)**. Customers experiencing longer wait times are more likely to cancel their bookings and have a poorer overall experience. This indicates the need to improve driver allocation, especially during peak demand periods and in high-traffic locations, to reduce waiting time and improve customer satisfaction and ride completion rates.

5. What is the lowest customer rating received, and what does it indicate about the overall customer experience?

A. Query :

```
select customer_id, min(customer_rating) as low_ratings_by_customers from bookings
```

```
where customer_rating is not null and customer_rating<4
```

```
group by customer_id order by low_ratings_by_customers asc limit 10;
```

Result :

	customer_id	low_ratings_by_customers
▶	CID885746	3
	CID494616	3
	CID111964	3
	CID259729	3
	CID693171	3
	CID246186	3

Result 17 ✕

Business Insight :

The analysis shows that the lowest customer rating recorded was **3 out of 5**, indicating that while no extremely poor ratings were received, there is still room to improve the customer experience. Analyzing the rides associated with lower ratings can help identify issues such as longer waiting times, ride cancellations, driver behavior, or service quality. Addressing these factors can improve customer satisfaction, encourage repeat bookings, and strengthen overall platform performance.

6. Which payment method is preferred by customers, and what does it reveal about customer payment behavior?

A. Query :

```
select customer_id, payment_method as digital_payments, count(booking_id) as total_bookings from bookings
```

```
where payment_method in ('upi', 'credit card')
```

```
group by customer_id, payment_method
```

```
order by total_bookings desc;
```

Result :

	customer_id	digital_payments	total_bookings
▶	CID288207	UPI	4
	CID221217	UPI	3
	CID472301	UPI	3
	CID838803	UPI	3
	CID406782	UPI	3
	CID683457	UPI	3

Business Insight :

The analysis indicates that **UPI is the most preferred digital payment method** among customers, reflecting a strong preference for fast, secure, and cashless transactions. This highlights the importance of maintaining a seamless UPI payment experience, offering UPI-based promotional incentives, and ensuring high payment

success rates to enhance customer convenience, encourage repeat bookings, and support revenue growth.

7. What is the average amount spent by each customer per booking, and what does it indicate about customer spending behavior?

A. Query :

```
SELECT
customer_id,
COUNT(booking_id) AS total_bookings,
SUM(booking_value) AS total_spent,
ROUND(
    SUM(booking_value) / NULLIF(COUNT(booking_id), 0),
    2
) AS avg_booking_value
FROM bookings
GROUP BY customer_id
ORDER BY
    total_spent DESC,
    total_bookings DESC,
    avg_booking_value DESC;
```

Result :

	customer_id	total_bookings	total_spent	avg_booking_value
▶	CID785112	3	8025	2675.00
	CID308763	3	6281	2093.67
	CID734557	3	6177	2059.00
	CID353074	3	6110	2036.67
	CID836942	4	6019	1504.75

Business Insight :

The average customer spending highlights the typical revenue generated per booking. Monitoring this metric helps identify customer spending patterns and supports pricing, promotional, and revenue optimization strategies.

8. How frequently do customers book rides, and how many days are they actively using the platform?

A. Query :

```
SELECT
customer_id,
COUNT(booking_id) AS total_bookings,
DATEDIFF(MAX(booking_date), MIN(booking_date)) AS active_days,
CASE
-- If active days is 0, just return their total booking count for that day
    WHEN DATEDIFF(MAX(booking_date), MIN(booking_date)) = 0 THEN
COUNT(booking_id)
    ELSE ROUND(COUNT(booking_id) / (DATEDIFF(MAX(booking_date),
MIN(booking_date)) / 30.0), 2)
END AS bookings_per_month
FROM
bookings
GROUP BY
customer_id order by bookings_per_month desc;
```

Result :

	customer_id	total_bookings	active_days	bookings_per_month
▶	CID981881	3	1	90.00
	CID925549	2	1	60.00
	CID715518	2	1	60.00
	CID904650	2	1	60.00
	CID582247	2	1	60.00
	CID411974	2	1	60.00
	CID333118	2	1	60.00

Business Insight :

Customer booking frequency and active days help identify loyal and highly engaged users. Customers with higher booking frequency and consistent activity are valuable for the business and can be targeted with loyalty programs and personalized offers to improve retention and maximize customer lifetime value.

9. Which customers have experienced repeated incomplete rides?

A. Query :

```
SELECT
customer_id,
COUNT(*) AS repeated_incomplete_rides
FROM bookings
WHERE incomplete_rides = 'Yes'
OR incomplete_rides IS NULL
GROUP BY customer_id
HAVING COUNT(*) > 1
ORDER BY repeated_incomplete_rides DESC;
```

Result :

	customer_id	repeated_incomplete_rides
▶	CID381415	4
	CID460299	4
	CID340854	4
	CID483311	3
	CID867762	3
	CID163765	3
	CID655004	2

Business Insight :

The analysis identifies customers with multiple incomplete rides, with some customers experiencing up to 4 incomplete rides. These recurring service failures may indicate operational issues such as driver cancellations or vehicle unavailability.

10. Which high-value customers have experienced poor service based on their booking frequency and average customer rating?**A. Query :**

```
select customer_id, count(booking_id) as total_bookings,
round(avg(customer_rating),2) as avg_customer_rating
from bookings group by customer_id having count(booking_id)>1 and
avg(customer_rating)=3 order by total_bookings desc;
```

Result :

	customer_id	total_bookings	avg_customer_rating
▶	CID888630	3	3
	CID584430	3	3
	CID503293	3	3
	CID379477	3	3
	CID639986	3	3
	CID939555	2	3

Business Insight :

The analysis identifies high-value customers with frequent bookings but lower average ratings. These customers are at a higher risk of churn and should be prioritized through personalized support, service improvements, and retention initiatives to preserve long-term revenue and customer loyalty.

11. Which high-value customers have spent the most on bookings but have a low average customer rating?

A. Query :

```
select customer_id, sum(booking_value) as total_spent, count(booking_id) as
total_bookings,
```

```
round(avg(customer_rating),2) as avg_customer_rating from bookings group by
customer_id having count(booking_id)>1 and
```

```
avg(customer_rating)=3
```

```
order by total_spent desc;
```

Result :

	customer_id	total_spent	total_bookings	avg_customer_rating
▶	CID807988	5004	2	3
	CID639986	3427	3	3
	CID888630	3125	3	3
	CID424697	3023	2	3
	CID931227	2765	2	3
	CID467820	2707	2	3

Business Insight :

The analysis identifies customers with the highest booking value but an average **rating of only 3**. These high-value customers are at risk of churn due to poor service experiences. Improving service quality and offering personalized support can help retain these valuable customers and protect future revenue.

Module 3 — Driver & Fleet Analytics

Business Questions

1. What is the average driver rating, and what does it indicate about overall driver performance?

A. Query :

```
select round(avg(driver_ratings),2) as avg_driver_rating from bookings;
```

Result :

	avg_driver_rating
▶	4

Business Insight :

The analysis shows an average driver **rating of 4.0/5**, indicating generally good service quality. Maintaining driver performance through regular training, performance monitoring, and customer feedback can further improve customer satisfaction and ride quality.

2. Driver cancellation analysis

A. Query :

```
SELECT

Canceled_Rides_by_Driver AS Cancellation_Reason,

COUNT(*) AS Total_Cancellations,

ROUND(COUNT(*) * 100.0 / SUM(COUNT(*) OVER(), 2) AS

Percentage_Of_Driver_Cancellations

FROM

bookings

WHERE

Canceled_Rides_by_Driver IS NOT NULL

AND Canceled_Rides_by_Driver <> 'null'

GROUP BY

Canceled_Rides_by_Driver

ORDER BY
```

Total_Cancellations DESC;

Result :

	Cancellation_Reason	Total_Cancellations	Percentage_Of_Driver_Cancellations
▶	Personal & Car related issue	6542	35.49
	Customer related issue	5413	29.36
	Customer was coughing/sick	3654	19.82
	More than permitted people in there	2825	15.32

Business Insight :

Driver cancellations are primarily caused by personal and **vehicle-related issues (35.49%), followed by customer-related issues (29.36%)**. Reducing these top cancellation reasons can improve ride completion rates, customer satisfaction, and overall platform reliability.

3. Vehicle-wise performance, Vehicle-wise revenue, Vehicle-wise completion rate.

A. Query :

```
select vehicle_type, count(booking_id) as total_bookings, sum(booking_value) as
total_value, round(avg(booking_value),2) as avg_value,
count(case when canceled_rides_by_driver is not null and canceled_rides_by_driver
<> 'null' then 1 end) as total_driver_cancellations,
round(count(case when canceled_rides_by_driver is not null and
canceled_rides_by_driver <> 'null' then 1 end)*100.0/count(*),2) as
driver_cancellation_rate
from bookings where vehicle_type is not null and vehicle_type<>'Null'
group by vehicle_type order by total_value desc;
```

Result :

	vehide_type	total_bookings	total_value	avg_value	total_driver_cancellations	driver_cancellation_rate
▶	Prime Sedan	14877	14877.2	557.81	2591	17.42
	eBike	14816	8181523	552.21	2694	18.18
	Auto	14755	8092210	548.44	2675	18.13
	Prime Plus	14707	8050889	547.42	2683	18.24
	Mini	14552	7990996	549.13	2584	17.76
	Bike	14662	7987054	544.75	2559	17.45
	Prime SUV	14655	7933330	541.34	2648	18.07

Business Insight :

Prime Sedan generated the highest revenue and bookings, while driver cancellation rates remained **fairly consistent (around 17–18%)** across all vehicle types, indicating similar operational performance regardless of vehicle category.

4. Vehicle-wise waiting time

A. Query :

```
select vehicle_type, avg(v_tat) as avg_waiting_time, max(v_tat) as  
maximum_waiting_time, min(v_tat) as minimum_waiting_time
```

```
from bookings
```

```
where vehicle_type is not null and vehicle_type <> 'Null' and v_tat is not null and  
v_tat <> 'Null'
```

```
group by vehicle_type order by avg_waiting_time desc;
```

Result :

	vehicle_type	avg_waiting_time	maximum_waiting_time	minimum_waiting_time
▶	Mini	172.6171	308	35
	Auto	171.6982	308	35
	Prime Plus	171.2104	308	35
	Bike	171.0724	308	35
	Prime Sedan	170.6421	308	35
	eBike	170.0680	308	35
	Prime SUV	168.8279	308	35

Business Insight :

Mini rides recorded the highest average waiting time at 172.61 seconds, indicating potential driver availability or allocation issues for this vehicle category. Reducing wait times can improve customer satisfaction and booking completion rates.

5. Vehicle-wise ride distance

A. Query :

```
select vehicle_type, avg(ride_distance) as avg_distance, max(ride_distance) as  
max_distance,
```

```
min(ride_distance) as min_distance
```

```
from bookings
```

```
group by vehicle_type order by avg_distance desc;
```

Result :

	vehicle_type	avg_distance	max_distance	min_distance
▶	Prime Sedan	15.7649	49	0
	eBike	15.5806	49	0
	Bike	15.5331	49	0
	Mini	15.5101	49	0
	Prime Plus	15.4475	49	0
	Prime SUV	15.2745	49	0
	Auto	6.2381	19	0

Business Insight :

Prime Sedan recorded the highest average ride distance of 15.76 km, indicating that customers prefer this vehicle category for longer trips, making it a key contributor to revenue generation.

6. Which are the top 5 vehicle types generating the highest revenue?

A. Query :

```
select vehicle_type, sum(booking_value) as total_revenue
from bookings group by vehicle_type order by total_revenue desc limit 5;
```

Result :

	vehicle_type	total_revenue
▶	Prime Sedan	8298612
	eBike	8181523
	Auto	8092210
	Prime Plus	8050889
	Mini	7990996

Business Insight :

Prime Sedan ranked first by generating ₹82,98,621 in total revenue, making it the highest revenue-contributing vehicle type. This suggests strong customer demand and highlights Prime Sedan as a key driver of business revenue.

7. Which vehicle type experienced the highest number of ride cancellations?

A. Query :

```
select vehicle_type, count(case when booking_status in
('Canceled by driver','Canceled by customer', 'Driver not found') then 1 end) as
highest_cancellations
from bookings where booking_status is not null and booking_status <> 'Null'
group by vehicle_type
```

order by highest_cancellations desc limit 5;

Result :

	vehicle_type	highest_cancellations
▶	Prime SUV	5659
	eBike	5636
	Prime Plus	5632
	Auto	5588
	Bike	5528

Business Insight :

Prime SUV recorded the highest number of cancellations with 5,659 rides, indicating potential issues related to driver availability, customer demand, or operational efficiency for this vehicle category. Reducing cancellations could improve ride completion and revenue.

8. Which vehicle type recorded the highest number of incomplete rides?

A. Query :

select vehicle_type, count(*) as highest_incomplete_rides

from bookings where incomplete_rides ='yes' or incomplete_rides is null

group by vehicle_type order by highest_incomplete_rides desc;

Result :

	vehicle_type	highest_incomplete_rides
▶	eBike	6210
	Prime SUV	6206
	Prime Plus	6168
	Auto	6150
	Prime Sedan	6109
	Bike	6107
	Mini	6033

Business Insight :

eBike recorded the highest number of incomplete rides with 6,210 cases, indicating potential operational or service delivery challenges. Addressing the causes of these incomplete rides can improve ride completion rates, customer satisfaction, and overall operational efficiency.

Module 4 — Operational Analytics

Business Questions

1. What is the peak booking hour, and how many bookings are recorded during that period?

A. Query :

```
select hour(time) as peak_booking_hour, count(*) as total_bookings from bookings  
group by peak_booking_hour order by total_bookings desc;
```

Result :

	peak_booking_hour	total_bookings
▶	12	4408
	15	4376
	8	4374
	17	4353
	9	4347
	21	4343
	10	4334
	1	4329
	0	4318
	2	4305

Business Insight :

The highest booking demand occurred at **12:00 AM, with 4,408 bookings**. This peak period highlights the need to ensure adequate driver availability and operational readiness during high-demand hours to reduce waiting times and improve ride completion rates.

2. Which day of the week records the highest number of bookings?

A. Query :

```
select dayname(booking_date) as peak_booking_weekdays, count(*) as  
total_bookings from bookings  
group by peak_booking_weekdays order by total_bookings desc;
```

Result :

	peak_booking_weekdays	total_bookings
▶	Tuesday	16926
	Monday	16729
	Wednesday	16434
	Friday	13345
	Saturday	13341
	Thursday	13236
	Sunday	13013

Business Insight :

Tuesday recorded the highest booking volume with 16,926 bookings, indicating strong mid-week demand. Increasing driver availability and optimizing operations on Tuesdays can help maximize ride completions and revenue.

3. Which pickup location generates the highest booking demand and revenue?

A. Query :

```
select pickup_location, count(booking_id) as total_bookings,
round(sum(booking_value),2) as total_booking_value
```

```
from bookings where pickup_location is not null and pickup_location <> 'Null'
```

```
group by pickup_location order by total_bookings desc limit 10;
```

Result :

	pickup_location	total_bookings	total_booking_value
▶	Banashankari	2201	1220409
	Yeshwanthpur	2139	1188522
	RT Nagar	2135	1225150
	Indiranagar	2133	1132068
	Sahakar Nagar	2126	1135877
	Basavanagudi	2120	1165243
	Ramamurthy Nagar	2116	1157016
	Vijayanagar	2113	1151424
	Tumkur Road	2105	1166372
	Cox Town	2100	1163196

Business Insight :

Banashankari emerged as the top pickup hotspot with **2,201 bookings** and a total booking value of **₹12,20,409**. This location represents a high-demand area, making it a priority for driver allocation and operational planning to improve service availability and maximize revenue.

4. Which drop location records the highest number of completed bookings and revenue?

A. Query :

```
select drop_location, count(booking_id) as total_bookings,  
round(sum(booking_value),2) as total_booking_value  
from bookings where drop_location is not null and drop_location <> 'Null'  
group by drop_location order by total_bookings desc limit 10;
```

Result :

	drop_location	total_bookings	total_booking_value
▶	Peenya	2159	1219900
	Mysore Road	2148	1204904
	MG Road	2128	1148577
	Hennur	2120	1129429
	HSR Layout	2117	1111477
	Sarjapur Road	2108	1115256
	Koramangala	2105	1153159
	Marathahalli	2104	1148542
	Vijayanagar	2103	1184720
	Hebbal	2097	1144728

Business Insight :

Peenya emerged as the top drop hotspot with 2,159 bookings and a total booking value of ₹12,19,900. This indicates consistently high demand at this destination, making it a key area for optimizing driver availability and improving operational efficiency.

5. Which pickup and drop location pairs are the most frequently traveled routes?

A. Query :

```
select pickup_location, drop_location, concat(pickup_location, ' -> ',drop_location)  
as popular_routes,  
count(booking_id) as total_bookings, round(sum(booking_value),2) as  
total_booking_value  
from bookings where pickup_location is not null and pickup_location <> 'null' and  
drop_location is not null and drop_location <> 'Null'  
group by pickup_location, drop_location order by total_bookings desc limit 10;
```

Result :

	pickup_location	drop_location	popular_routes	total_bookings	total_booking_value
►	Majestic	Hennur	Majestic -> Hennur	67	28710
	Ramamurthy Nagar	Frazer Town	Ramamurthy Nagar -> Frazer Town	62	34006
	Sahakar Nagar	Yeshwanthpur	Sahakar Nagar -> Yeshwanthpur	62	32215
	Kammanahalli	Kadugodi	Kammanahalli -> Kadugodi	61	28360
	Magadi Road	Vijayanagar	Magadi Road -> Vijayanagar	60	40745
	Bellandur	Indiranagar	Bellandur -> Indiranagar	60	35258
	Nagarbhavi	Nagarbhavi	Nagarbhavi -> Nagarbhavi	60	32487
	Hulimavu	Banashankari	Hulimavu -> Banashankari	60	37022
	KR Puram	Electronic City	KR Puram -> Electronic City	59	31130
	Magadi Road	Electronic City	Magadi Road -> Electronic City	59	29019

Business Insight :

Majestic → Hennur emerged as the most popular route with 67 bookings, indicating a consistently **high travel demand**. Such high-demand routes can be prioritized for better driver availability and optimized fleet allocation to reduce passenger waiting time and improve service efficiency.

6. Which pickup-to-drop routes have the highest average ride distance?

A. Query :

```
select pickup_location, drop_location, concat(pickup_location,' -> ',drop_location)
as route_name,

count(booking_id) as total_bookings, round(avg(ride_distance),2) as
avg_ride_distance

from bookings where

pickup_location is not null and pickup_location <> 'Null' and drop_location is not
null and drop_location <> 'Null'

and ride_distance is not null and ride_distance <> 'Null' and ride_distance>0

group by pickup_location, drop_location

order by avg_ride_distance desc limit 10;
```

Result :

	pickup_location	drop_location	route_name	total_bookings	avg_ride_distance
►	Devanahalli	Banashankari	Devanahalli -> Banashankari	16	32.63
	Jayanagar	Richmond Town	Jayanagar -> Richmond Town	17	32.29
	Nagarbhavi	Jayanagar	Nagarbhavi -> Yelahanka	19	32.26
	Hennur	Majestic	Hennur -> Majestic	21	31.95
	Sarjapur Road	Kadugodi	Sarjapur Road -> Kadugodi	15	31.93
	Tumkur Road	Richmond Town	Tumkur Road -> Richmond Town	15	31.80
	KR Puram	Majestic	KR Puram -> Majestic	10	30.90
	Peenya	Ulsoor	Peenya -> Ulsoor	20	30.90
	Yeshwanthpur	Electronic City	Yeshwanthpur -> Electronic City	32	30.78
	Hennur	Sahakar Nagar	Hennur -> Sahakar Nagar	25	30.64

Business Insight :

Devanahalli → Banashankari recorded the highest average ride distance (32.63 km), indicating demand for long-distance travel. Such routes can be prioritized by assigning suitable vehicles and ensuring driver availability to improve customer experience and maximize revenue from premium rides.

7. Average V_TAT, C_TAT

A. Query :

```
select vehicle_type, round(avg(v_tat),2) as avg_v_tat,
round(avg(c_tat),2) as avg_c_tat from bookings group by vehicle_type order by
avg_v_tat desc;
```

Result :

	vehicle_type	avg_v_tat	avg_c_tat
►	Mini	172.62	84.45
	Auto	171.70	85.34
	Prime Plus	171.21	84.75
	Bike	171.07	85.12
	Prime Sedan	170.64	85.40
	eBike	170.07	84.98
	Prime SUV	168.83	84.05

Business Insight :

VTAT (Vehicle Turnaround Time) = 172.62s is much higher than CTAT = 84.45s, meaning vehicles take **~2x more time to complete turnaround** compared to customer turnaround, indicating delays in vehicle processing/operations flow.

8. Which pickup locations have the highest bookings and longest waiting time impact?

A. Query :

select pickup_location, count(booking_id) as total_bookings, round(avg(v_tat),2) as avg_waiting_time from bookings

where pickup_location is not null and pickup_location <> 'Null' and v_tat is not null and v_tat <> 'Null'

group by pickup_location order by avg_waiting_time desc limit 10;

Result :

	pickup_location	total_bookings	avg_waiting_time
►	Varthur	1270	180.32
	Yeshwanthpur	1353	175.05
	Vijayanagar	1264	174.96
	Ulsoor	1290	174.22
	Hebbal	1288	174.16
	Electronic City	1290	173.87
	Frazer Town	1277	173.81
	Bellandur	1250	173.44
	Majestic	1287	173.29
	Richmond Town	1275	173.22

Business Insight :

Varthur leads with 1270 total bookings and a high avg waiting time of 180.32s, showing it is a high-demand hotspot with potential capacity or supply issues causing longer delays

9. How do operational delays differ in mini across booking flow stages?

A. Query :

select vehicle_type, count(booking_id) as total_bookings,

round(avg(v_tat),2) as driver_arrival_delay,

round(avg(c_tat),2) as customer_arrival_delay from bookings

where v_tat is not null and v_tat <> 'Null' and c_tat is not null and c_tat <> 'Null'

and vehicle_type is not null and vehicle_type <> 'Null'

group by vehicle_type

order by driver_arrival_delay desc;

Result :

	vehicle_type	total_bookings	driver_arrival_delay	customer_arrival_delay
▶	Mini	9036	172.62	84.45
	Auto	9167	171.70	85.34
	Prime Plus	9075	171.21	84.75
	Bike	9134	171.07	85.12
	Prime Sedan	9379	170.64	85.40
	eBike	9180	170.07	84.98
	Prime SUV	8996	168.83	84.05

Business Insight :

In mini, driver arrival delay (172.62s) is much higher than customer arrival delay (84.45s) for 9036 bookings, indicating the main bottleneck is on the driver side rather than customers.

Module 5 — Revenue Analytics

Business Questions

1. Which revenue segment contributes the most to total revenue?

A. Query :

with revenue_segments as(

select booking_id, booking_value,

case when booking_value between 0 and 100 then 'Low Fare'

when booking_value between 101 and 400 then 'Mid fare'

when booking_value between 401 and 700 then 'High fare'

else '700+ luxury ride' end as revenue_bucket from bookings

where booking_value is not null and booking_value <> 'Null'

)

select revenue_bucket, count(booking_id) as total_bookings,

round(sum(booking_value),2) as total_revenue

from revenue_segments group by revenue_bucket order by total_revenue desc;

Result :

	revenue_bucket	total_bookings	total_revenue
▶	700+ luxury ride	22307	29758377
	Mid fare	53986	13500681
	High fare	26571	13259556
	Low Fare	160	16000

Business Insight :

Mid Fare generates the **highest revenue (13.5M) from 53,986 bookings**, making it the primary revenue driver despite the availability of higher-priced ride segments.

2. Which vehicle type generates the highest revenue?

A. Query :

with vehicle_revenue as(

select vehicle_type, booking_id, booking_value from bookings where

vehicle_type is not null and vehicle_type <> 'Null'

and booking_value is not null and booking_value <> 'Null'

and booking_value > 0

)

select vehicle_type, count(booking_id) as total_bookings,

round(sum(booking_value),2) as revenue_by_vehicle from vehicle_revenue

group by vehicle_type order by revenue_by_vehicle desc;

Result :

	vehicle_type	total_bookings	revenue_by_vehicle
▶	Prime Sedan	14877	8298612
	eBike	14816	8181523
	Auto	14755	8092210
	Prime Plus	14707	8050889
	Mini	14552	7990996
	Bike	14662	7987054
	Prime SUV	14655	7933330

Business Insight :

Prime Sedan ranks first with **14,877 bookings and ₹82,98,612 in revenue**, making it the top revenue-generating vehicle type.

3. Revenue by payment method

A. Query :

with method_wise_revenue as (

select vehicle_type, booking_id, booking_value, payment_method from bookings
where payment_method is not null and payment_method <> 'Null' and

booking_value is not null and booking_value <> 'Null' and booking_value > 0

)

```
select vehicle_type,payment_method, count(booking_id) as total_bookings,  
round(sum(booking_value),2) as revenue_by_payment_method
```

```
from method_wise_revenue group by vehicle_type, payment_method order by  
revenue_by_payment_method desc;
```

Result :

	vehide_type	payment_method	total_bookings	revenue_by_payment_method
▶	Prime Sedan	Cash	5105	2820574
	Prime Plus	Cash	5000	2778957
	Auto	Cash	5031	2778195
	Prime SUV	Cash	4987	2753830
	Bike	Cash	4993	2742917
	eBike	Cash	5006	2741825
	Mini	Cash	4900	2644018

Business Insight :

Cash contributes the highest revenue, indicating it is the most preferred payment method among customers.

4. Which pickup locations generate the highest revenue?

A. Query :

```
with revenue_calculation as (
```

```
    select vehicle_type, booking_id, pickup_location, booking_value from bookings
```

```
    where vehicle_type is not null and vehicle_type <> 'Null'
```

```
    and pickup_location is not null and pickup_location <> 'Null'
```

```
    and booking_value is not null and booking_value <> 'Null'
```

```
    and booking_value > 0
```

```
)
```

```
select vehicle_type, count(booking_id) as total_bookings, pickup_location,  
round(sum(booking_value),2) as total_revenue
```

```
from revenue_calculation group by vehicle_type, pickup_location order by  
total_revenue desc limit 10;
```

Result :

	vehide_type	total_bookings	pickup_location	total_revenue
►	Prime Sedan	339	Peenya	211452
	Prime Sedan	296	Nagarbhavi	200178
	Auto	360	Banashankari	198073
	Bike	323	Yeshwanthpur	197736
	eBike	316	RT Nagar	195726
	Auto	340	BTM Layout	195560
	Mini	321	Marathahalli	194816
	Auto	325	RT Nagar	190291
	Prime Sedan	321	Hebbal	188748
	eBike	293	Ramamurthy Nagar	186849

Business Insight :

The top 10 pickup locations contribute the majority of revenue, highlighting key demand hotspots where maintaining adequate driver supply can maximize bookings and revenue.

5. Which drop locations and vehicle types generate the highest revenue?

A. Query :

with revenue_drop as (

select vehicle_type, booking_id, drop_location, booking_value from bookings

where vehicle_type is not null and vehicle_type <> 'Null'

and drop_location is not null and drop_location <> ' Null'

and booking_value is not null and booking_value <> 'Null'

and booking_value > 0

)

select vehicle_type, count(booking_id) as total_bookings, drop_location,

round(sum(booking_value),2) as total_revenue

from revenue_drop group by vehicle_type, drop_location order by total_revenue desc;

Result :

	vehide_type	total_bookings	drop_location	total_revenue
►	Bike	302	Shantinagar	203719
	eBike	318	Magadi Road	194486
	Mini	315	BTM Layout	194080
	Prime Sedan	312	Hulimavu	192570
	Mini	316	Devanahalli	190946
	Prime Sedan	313	Malleshwaram	188759
	Prime Sedan	305	Rajarajeshwari Nagar	188410
	Prime SUV	315	Basavanagudi	188115

Business Insight :

The top-performing drop location-vehicle type combinations contribute the highest revenue, helping identify the most profitable routes and vehicle categories for better fleet allocation and demand planning.

6. Which weekday generates the highest revenue?

A. Query :

with weekday_revenue as(

select booking_date, dayname(booking_date) as weekday_name,

dayofweek(booking_date) as weekday_index, booking_id, booking_value

from bookings where booking_value is not null and booking_value <> 'Null'

and booking_value > 0

)

select weekday_name, count(booking_id) as total_bookings,

round(sum(booking_value),2) as total_revenue

from weekday_revenue group by weekday_name, weekday_index order by
total_revenue desc;

Result :

	weekday_name	total_bookings	total_revenue
►	Tuesday	16926	9311131
	Monday	16729	9273756
	Wednesday	16434	9040761
	Thursday	13236	7354322
	Saturday	13341	7301461
	Friday	13345	7201319
	Sunday	13013	7051864

Business Insight :

Tuesday generates the highest revenue, indicating it is the strongest revenue-driving day of the week and a key period for optimizing driver availability and operations.

7. Which booking hour generates the highest revenue?

A. Query :

with revenue_hour as(

select hour(time) as booking_hour, booking_id, booking_value from bookings

where booking_value is not null and booking_value <> 'Null'

```

    and booking_value > 0
)

select booking_hour, count(booking_id) as total_bookings,
round(sum(booking_value),2) as total_revenue

from revenue_hour group by booking_hour order by total_revenue desc;

```

Result :

	booking_hour	total_bookings	total_revenue
▶	13	4285	2444769
	7	4304	2444489
	12	4408	2436926
	1	4329	2416576
	0	4318	2399829
	6	4271	2387209
	21	4343	2384950
	2	4305	2368167
	10	4334	2364817
	9	4347	2359404

Business Insight :

1 PM (13:00) generates the highest revenue (₹24,44,769) with 4,285 bookings, making it the peak revenue hour and an ideal time to ensure maximum driver availability.

8. How much revenue is lost due to customer cancellations?

A. Query :

```

with revenue_lost as (

    select booking_id, canceled_rides_by_customer, booking_value from bookings

    where canceled_rides_by_customer is not null and canceled_rides_by_customer
    <> 'Null'

    and booking_value is not null and booking_value <> 'Null' and booking_value > 0
)

select count(booking_id) as total_bookings, count(canceled_rides_by_customer) as
customer_cancellations,

round(sum(booking_value),2) as total_revenue_lost from revenue_lost;

```

Result :

	total_bookings	customer_cancellations	total_revenue_lost
▶	10499	10499	5770901

Business Insight :

Customer cancellations caused a revenue loss of **₹57,70,901 (₹57.71 lakh)**, indicating a significant impact on overall business revenue.

9. How much revenue is lost due to driver cancellations?

A. Query :

with revenue_lost_from_driver as(

select canceled_rides_by_driver, booking_value from bookings

where canceled_rides_by_driver is not null and canceled_rides_by_driver <> 'Null'

and booking_value is not null and booking_value <> 'Null'

and booking_value > 0

)

select count(canceled_rides_by_driver) as driver_cancellations,
round(sum(booking_value),2) as total_revenue_lost

from revenue_lost_from_driver;

Result :

	driver_cancellations	total_revenue_lost
▶	18434	10183427

Business Insight :

Driver cancellations resulted in a revenue loss of ₹1,01,83,427 (₹1.02 crore), indicating a major impact on overall business revenue and the need to improve driver reliability.

10. How much revenue is lost due to incomplete rides?

A. Query :

with revenue_lost_by_incomplete_rides as (

select incomplete_rides, booking_value from bookings

where incomplete_rides != 'No' and booking_value is not null and booking_value <> 'Null'


```

        and booking_value > 0
    )

select count(incomplete_rides) as incomplete_rides_count,
round(sum(booking_value),2) as total_revenue_lost_by_incomplete_rides
from revenue_lost_by_incomplete_rides;

```

Result :

	incomplete_rides_count	total_revenue_lost_by_incomplete_rides
▶	3926	2250482

Business Insight :

Incomplete rides resulted in a revenue loss of ₹22,50,482 (₹22.50 lakh), highlighting the need to minimize ride failures and improve ride completion rates.

9. Advanced SQL

Window Functions

- **Rank top 10 pickup locations by revenue.**

A. Query :

```

with top_locations as(
    select pickup_location, booking_id, booking_value from bookings
    where pickup_location is not null and pickup_location <> 'Null'
    and booking_value is not null and booking_value <> 'Null'
    and booking_value > 0
)
select pickup_location, count(booking_id) as total_bookings,
round(sum(booking_value),2) as total_revenue,
dense_rank() over(order by count(booking_id) desc) as location_rank
from top_locations group by pickup_location order by total_revenue desc limit
10;

```

Result :

	pickup_location	total_bookings	total_revenue	location_rank
▶	RT Nagar	2135	1225150	3
	Banashankari	2201	1220409	1
	Nagarbhavi	2083	1211057	16
	Yeshwanthpur	2139	1188522	2
	Frazer Town	2048	1175684	25
	Hosur Road	2074	1168154	18
	Tumkur Road	2105	1166372	9
	Basavanagudi	2120	1165243	6
	Cox Town	2100	1163196	10
	JP Nagar	2039	1159478	29

- Find the top 3 busiest pickup locations.

A. Query :

with busy_locations as(

select booking_id, pickup_location from bookings where pickup_location is not null and pickup_location <> 'Null'

)

select pickup_location, count(booking_id) as total_bookings, dense_rank()

over(order by count(booking_id) desc) as location_rank

from busy_locations group by pickup_location order by total_bookings desc limit 3;

Result :

	pickup_location	total_bookings	location_rank
▶	Banashankari	2201	1
	Yeshwanthpur	2139	2
	RT Nagar	2135	3

- Identify the top 5 customers by booking value.

A. Query :

with top_customers as (

select customer_id, booking_id, booking_value from bookings

where booking_value is not null and booking_value <> 'Null'

and booking_value > 0

)

select customer_id, count(booking_id) as total_bookings,

round(sum(booking_value),2) as total_revenue,

dense_rank() over(order by round(sum(booking_value),2) desc) as

customer_rank from top_customers

group by customer_id order by total_revenue desc limit 5;

Result :

	customer_id	total_bookings	total_revenue	customer_rank
▶	CID785112	3	8025	1
	CID308763	3	6281	2
	CID734557	3	6177	3
	CID353074	3	6110	4
	CID836942	4	6019	5

CTE

- **Compare current-day bookings with the previous day's bookings.**

A. Query :

```
with bookings_comparision as(
    select date(booking_date) as drive_date,
           count(booking_id) as total_bookings,
           round(sum(booking_value),2) as total_revenue from bookings
    where booking_date is not null
    group by date(booking_date)
)
select drive_date, total_bookings as current_day_bookings,
       lag(total_bookings,1) over(order by drive_date) as previous_day_bookings,
       (total_bookings-lag(total_bookings,1) over(order by drive_date)) as
       booking_difference,
       total_revenue as current_day_revenue, lag(total_revenue,1) over(order by
       drive_date) as previous_day_revenue
from bookings_comparision order by drive_date desc;
```

Result :

	drive_date	current_day_bookings	previous_day_bookings	booking_difference	current_day_revenue	previous_day_revenue
▶	2024-07-31	3072	3432	-360	1708838	1896548
	2024-07-30	3432	3370	62	1896548	1924027
	2024-07-29	3370	3246	124	1924027	1782082
	2024-07-28	3246	3362	-116	1782082	1837516
	2024-07-27	3362	3338	24	1837516	1866231
	2024-07-26	3338	3361	-23	1866231	1875293
	2024-07-25	3361	3286	75	1875293	1789575
	2024-07-24	3286	3377	-91	1789575	1832123
	2024-07-23	3377	3377	0	1832123	1812812
	2024-07-22	3377	3237	140	1812812	1752365

- **Identify vehicle types with above-average completion rates.**

A. Query :

```
with above_avg_vehicle_types as (
```

```

select vehicle_type,
count(case when incomplete_rides = 'No' then 1 end) as completion_ride,
count(booking_id) as total_bookings,
count(case when incomplete_rides = 'No' then 1
end)*100.0/count(booking_id) as completion_rate
from bookings where vehicle_type is not null and vehicle_type <> 'Null'
group by vehicle_type
)
select vehicle_type, round(completion_rate,2) as vehicle_completion_rate from
above_avg_vehicle_types
where completion_rate > (
select count(case when incomplete_rides = 'No' then 1
end)*100.0/count(booking_id) from bookings
)
order by completion_rate desc;

```

Result :

	vehicle_type	vehicle_completion_rate
▶	Prime Sedan	58.94
	Mini	58.54
	Bike	58.35
	Auto	58.32

Correlated Subqueries

- **Find top 10 customers whose average booking value is above the average of all customers using the same vehicle type.**

A. Query :

```

with avg_booking_value as (
select customer_id, vehicle_type, avg(booking_value) as
customer_avg_booking
from bookings where booking_value is not null and booking_value <> 'Null'
and booking_value > 0
group by customer_id, vehicle_type
),
vehicle_benchmark as(
select customer_id, vehicle_type, customer_avg_booking,
avg(customer_avg_booking) over(partition by vehicle_type) as
overall_vehicle_avg
from avg_booking_value
)

```

```
select customer_id, vehicle_type, round(customer_avg_booking,2) as
avg_booking_individual from vehicle_benchmark
where customer_avg_booking > overall_vehicle_avg order by
customer_avg_booking desc limit 10;
```

Result :

	customer_id	vehicle_type	avg_booking_individual
▶	CID520492	Prime SUV	Prime SUV
	CID449284	Mini	2999.00
	CID536592	Prime Sedan	2999.00
	CID159544	Prime Sedan	2998.00
	CID602417	Prime SUV	2998.00
	CID884826	Prime SUV	2998.00
	CID315200	Prime Plus	2998.00
	CID599856	eBike	2998.00
	CID980150	eBike	2998.00
	CID573552	eBike	2998.00

- **Find pickup locations whose cancellation rate is above the overall cancellation rate.**

A. Query :

```
with location_based as(
    select pickup_location, count(booking_id) as total_bookings, sum(case when
        canceled_rides_by_driver is not null or canceled_rides_by_customer is
        not null then 1 else 0 end) as cancelled_bookings,
        avg(booking_value) as avg_booking_value from bookings group by
        pickup_location
    )
select pickup_location, round(cancelled_bookings*100.0/total_bookings,2) as
cancellation_rate from location_based where
round(cancelled_bookings*100.0/total_bookings,2)> (
    select count(case when canceled_rides_by_driver is not null or
        canceled_rides_by_customer is not null then 1 end)/count(*) from bookings
);
```

Result :

	pickup_location	cancellation_rate
▶	Tumkur Road	30.40
	Magadi Road	28.49
	Sahakar Nagar	28.22
	HSR Layout	27.13
	Rajajinagar	27.82
	Kadugodi	26.77
	Bannerghatta Road	28.64
	Chamarajpet	29.08

Route Analysis

- Find routes with the highest average booking value.

A. Query :

```
select
    pickup_location,
    drop_location,
    concat(pickup_location, ' -> ', drop_location) as route_path,
    count(booking_id) as total_bookings,
    round(avg(booking_value), 2) as avg_booking_value
from bookings
where pickup_location is not null and pickup_location <> 'Null'
    and drop_location is not null and drop_location <> 'Null'
    and booking_value is not null and booking_value <> 'Null' and booking_value > 0
group by pickup_location, drop_location
order by avg_booking_value desc
limit 10;
```

Result :

	pickup_location	drop_location	route_path	total_bookings	avg_booking_value
►	JP Nagar	Hulimavu	JP Nagar -> Hulimavu	34	960.62
	Jayanagar	Hebbal	Jayanagar -> Hebbal	39	914.05
	Nagarbhavi	Hulimavu	Nagarbhavi -> Hulimavu	33	873.85
	Chamarajpet	Mysore Road	Chamarajpet -> Mysore Road	26	860.69
	Richmond Town	Koramangala	Richmond Town -> Koramangala	42	860.24
	Chickpet	Kadugodi	Chickpet -> Kadugodi	34	851.44
	Nagarbhavi	Kadugodi	Nagarbhavi -> Kadugodi	36	838.39
	Frazer Town	Shantinagar	Frazer Town -> Shantinagar	30	835.33
	HSR Layout	RT Nagar	HSR Layout -> RT Nagar	33	831.73
	HSR Layout	Ramamurthy Nagar	HSR Layout -> Ramamurthy Nagar	27	830.74

- Find routes with the highest operational delays.

A. Query :

```
select
    pickup_location,
    drop_location,
    concat(pickup_location, ' -> ', drop_location) as route_path,
    count(booking_id) as total_rides,
    round(avg(v_tat), 2) as avg_driver_delay,
    round(avg(c_tat), 2) as avg_customer_delay,
    round(avg(v_tat + c_tat), 2) as avg_total_operational_delay
```

from bookings

where pickup_location is not null and pickup_location <> 'Null'

and drop_location is not null and drop_location <> 'Null'

and (v_tat is not null or c_tat is not null)

group by pickup_location, drop_location

order by avg_total_operational_delay desc

limit 10;

Result :

	pickup_location	drop_location	route_path	total_rides	avg_driver_delay	avg_customer_delay	avg_total_operational_delay
►	Varthur	KR Puram	Varthur -> KR Puram	26	236.12	91.35	327.46
	Hulimavu	Yelahanka	Hulimavu -> Yelahanka	16	224.00	95.31	319.31
	Malleshwaram	HSR Layout	Malleshwaram -> HSR Layout	23	226.13	90.87	317.00
	Bannerghatta Road	Kadugodi	Bannerghatta Road -> Kadugodi	33	221.03	91.82	312.85
	Whitefield	Whitefield	Whitefield -> Whitefield	17	209.59	103.24	312.82
	Yeshwanthpur	Peenya	Yeshwanthpur -> Peenya	30	211.87	100.83	312.70
	Marathahalli	Marathahalli	Marathahalli -> Marathahalli	17	220.71	88.53	309.24
	Koramangala	BTM Layout	Koramangala -> BTM Layout	19	212.95	95.53	308.47
	HSR Layout	Rajajinagar	HSR Layout -> Rajajinagar	16	208.69	99.69	308.38
	Vijayanagar	Bannerghatta Road	Vijayanagar -> Bannerghatta Road	29	210.97	96.38	307.34

Time-Based Analysis

- **Identify peak booking hours for each vehicle type.**

A. Query :

```
with hourly_booking_metrics as (
  select
    vehicle_type,
    hour(time) as booking_hour,
    count(booking_id) as total_bookings,
    row_number() over(partition by vehicle_type order by count(booking_id)
desc) as rank_order
  from bookings
  where vehicle_type is not null and vehicle_type <> 'Null'
    and time is not null and time <> 'Null'
  group by vehicle_type, hour(time)
)
select
  vehicle_type,
  booking_hour as peak_hour,
  total_bookings as bookings_at_peak_hour
from hourly_booking_metrics
where rank_order = 1
order by total_bookings desc;
```

Result :

	vehicle_type	peak_hour	bookings_at_peak_hour
▶	eBike	21	668
	Prime SUV	9	663
	Prime Sedan	12	660
	Auto	17	658
	Mini	1	652
	Bike	0	649
	Prime Plus	11	642

- **Compare weekday vs. weekend performance.**

A. Query :

```

with day_performance_metrics as (
  select
    booking_id,
    booking_value,
    booking_status,
    dayofweek(time) as day_num,
    case when dayofweek(time) in (1, 7) then 'Weekend' else 'Weekday' end as
day_type
  from bookings
  where time is not null and time <> 'Null'
)
select
  day_type,
  count(booking_id) as total_bookings,
  round(avg(case when booking_value is not null and booking_value <> 'Null'
then booking_value else 0 end), 2) as avg_booking_value,
  round(sum(case when booking_value is not null and booking_value <> 'Null'
then booking_value else 0 end), 2) as total_revenue,
  round(
    sum(case when booking_status = 'Success' then 1 else 0 end) * 100.0
    / count(booking_id), 2
  ) as completion_rate_pct,
  round(
    sum(case when booking_status in( 'canceled by driver', 'canceled by
customer','driver not found') then 1 else 0 end) * 100.0
    / count(booking_id), 2
  ) as cancellation_rate_pct
from day_performance_metrics
group by day_type;

```


Result :

	day_type	total_bookings	avg_booking_value	total_revenue	completion_rate_pct	cancellation_rate_pct
►	Weekday	103024	548.75	56534614	62.09	37.91

Operational Performance

- Identify locations with consistently high V_TAT and C_TAT.

A. Query :

```
with global_tat_benchmark as (  
  select  
    avg(v_tat) as global_avg_v_tat,  
    avg(c_tat) as global_avg_c_tat  
  from bookings  
  where v_tat is not null and c_tat is not null  
)  
select  
  pickup_location,  
  count(booking_id) as total_rides_managed,  
  round(avg(v_tat), 2) as avg_driver_delay_vtat,  
  round(avg(c_tat), 2) as avg_customer_delay_ctat,  
  round(avg(v_tat + c_tat), 2) as total_combined_delay  
from bookings, global_tat_benchmark  
where pickup_location is not null and pickup_location <> 'Null'  
group by pickup_location  
having count(booking_id) >= 50  
  and avg(v_tat) > max(global_avg_v_tat)  
  and avg(c_tat) > max(global_avg_c_tat)  
order by total_combined_delay desc;
```

Result :

	pickup_location	total_rides_managed	avg_driver_delay_vtat	avg_customer_delay_ctat	total_combined_delay
►	Majestic	2067	173.29	87.23	260.52
	Vijayanagar	2113	174.96	85.53	260.49
	Hebbal	2058	174.16	86.14	260.30
	Richmond Town	2042	173.22	85.96	259.17
	Jayanagar	1985	173.05	86.09	259.13
	Bannerghatta Road	2046	171.77	87.34	259.10
	Kammanahalli	2071	172.80	85.96	258.76
	Kadugodi	2002	173.05	85.60	258.65
	Chamrajpet	1981	173.87	85.58	259.45

- **Compare operational efficiency across vehicle categories.**

A. Query :

```
select
  vehicle_type,
  count(booking_id) as total_bookings,
  round(
    sum(case when booking_status = 'Success' or (canceled_rides_by_driver is
null and canceled_rides_by_customer is null and incomplete_rides = 'No') then 1
else 0 end) * 100.0
    / count(booking_id), 2
  ) as completion_rate_pct,
  round(
    sum(case when canceled_rides_by_driver is not null or
canceled_rides_by_customer is not null or booking_status like '%Cancel%' then
1 else 0 end) * 100.0
    / count(booking_id), 2
  ) as cancellation_rate_pct,
  round(
    sum(case when (booking_status like '%Incomplete%' or incomplete_rides =
'Yes') and (canceled_rides_by_driver is null and canceled_rides_by_customer is
null) then 1 else 0 end) * 100.0
    / count(booking_id), 2
  ) as incomplete_rate_pct,
  round(avg(case when v_tat > 0 then v_tat else null end), 2) as
avg_driver_tat_delay
from bookings
where vehicle_type is not null and vehicle_type <> 'Null'
group by vehicle_type
order by total_bookings desc;
```

Result :

	vehicle_type	total_bookings	completion_rate_pct	cancellation_rate_pct	incomplete_rate_pct	avg_driver_tat_delay
►	Prime Sedan	14877	63.04	27.30	4.11	170.64
	eBike	14816	61.96	28.39	3.87	170.07
	Auto	14755	62.13	28.32	3.81	171.70
	Prime Plus	14707	61.71	28.27	3.64	171.21
	Bike	14662	62.30	27.94	3.95	171.07
	Prime SUV	14655	61.39	28.29	3.73	168.83
	Mini	14552	62.09	28.10	3.55	172.62

10. Power BI Dashboard Design

Dashboard 1 — Executive Overview

Audience: CXOs, Business Heads

KPI Cards

- Total Bookings
- Successful Bookings
- Ride Completion Rate
- Total Revenue
- Average Booking Value
- Cancellation Rate

Charts

- Booking Trend
- Booking Status Breakdown
- Vehicle Type Distribution

Filters

- Date
- Vehicle Type
- Payment Method



Dashboard 2 — Customer Insights

Audience: Customer Experience Team

KPI Cards

- Average Customer Rating
- Customer Waiting Time
- Customer Cancellation Rate

Visuals

- Customer Rating Distribution
- Top Customers
- Customer Cancellation Analysis
- Preferred Payment Method



Dashboard 3 — Driver & Fleet Performance

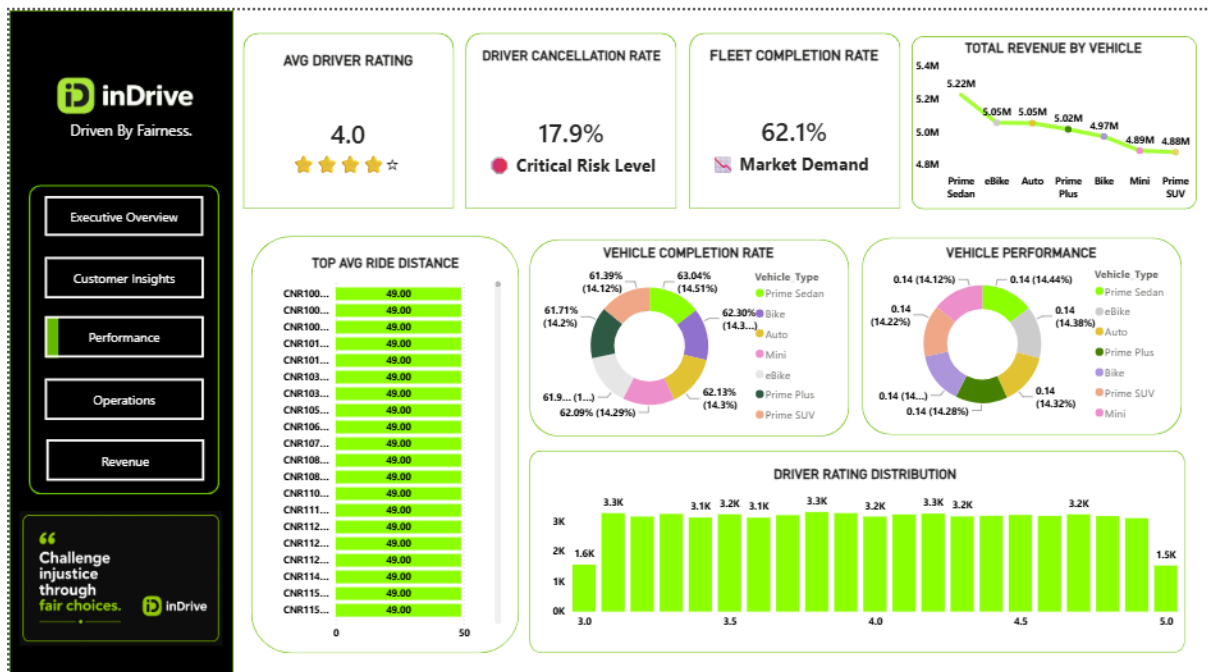
Audience: Fleet Operations

KPI Cards

- Average Driver Rating
- Driver Cancellation Rate
- Fleet Completion Rate

Visuals

- Driver Rating Distribution
- Vehicle Performance
- Vehicle Completion Rate
- Vehicle Revenue
- Avg Ride Distance Analysis



Dashboard 4 — Operations Dashboard

Audience: Operations Team

KPI Cards

- Average V_TAT
- Average C_TAT

Visuals

- Route Performance
- Peak Booking Hours
- Peak Booking Days



Dashboard 5 — Revenue Dashboard

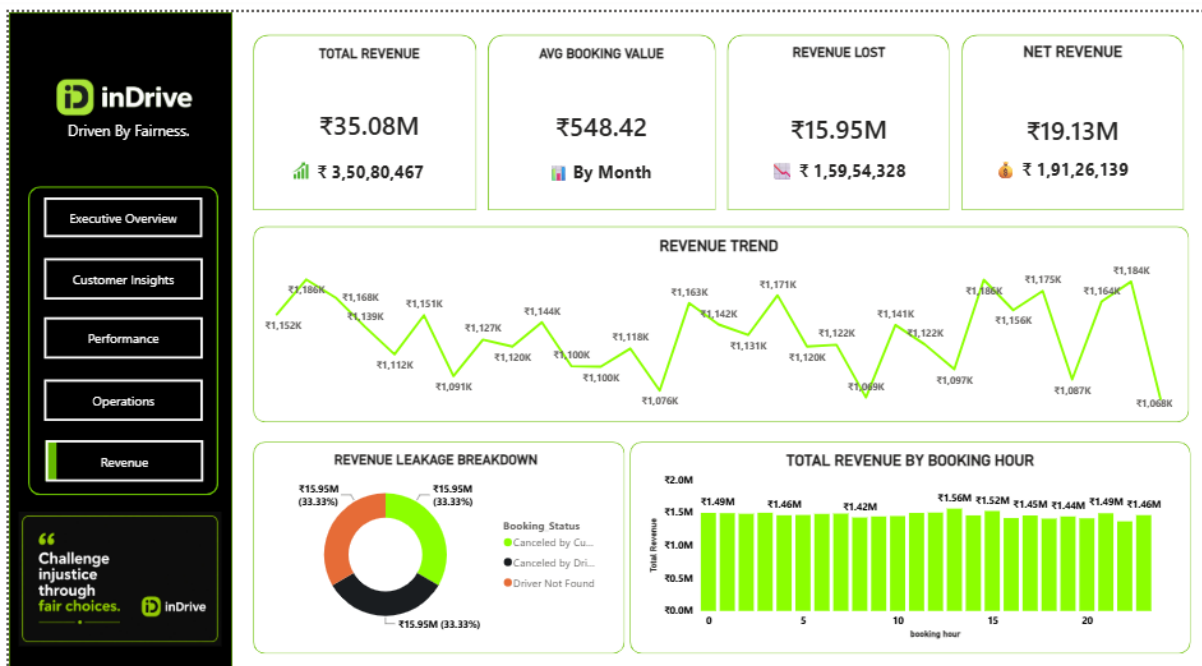
Audience: Finance & Strategy

KPI Cards

- Total Revenue
- Revenue Lost
- Average Booking Value

Visuals

- Revenue Trend
- Revenue by Pickup Location
- Revenue by Booking Hour
- Revenue Leakage Breakdown



11. Business Recommendations

1. Improve Ride Completion Rate

The current ride completion rate is 62.09%, indicating that a significant number of bookings do not result in successful rides. The company should investigate the primary causes of cancellations and failed bookings to improve operational efficiency and customer satisfaction.

2. Reduce Driver Cancellations

Driver cancellations represent a substantial share of unsuccessful bookings. Introducing incentive programs during peak demand periods, optimizing driver allocation, and monitoring driver performance can help reduce cancellation rates.

3. Strengthen Customer Retention

Approximately 38% of bookings remain incomplete due to cancellations or unavailable drivers. Reducing these unsuccessful bookings can improve customer trust and encourage repeat usage of the platform.

4. Optimize Fleet Allocation

Vehicle demand appears to be relatively balanced across different vehicle categories. Regular monitoring of vehicle utilization should continue to ensure adequate driver availability for each category while avoiding oversupply.

5. Monitor Daily Booking Trends

The daily booking trend remains relatively stable throughout the month with a few noticeable fluctuations. Further analysis should be conducted on low-demand days to identify external factors such as holidays, weather conditions, or operational constraints.

6. Increase Revenue Through Operational Efficiency

The platform generated ₹35.08 million in revenue from over 103,000 bookings. Improving the booking completion rate can directly increase realized revenue without requiring a proportional increase in customer acquisition.

7. Establish Executive KPIs

Management should continuously monitor key performance indicators including :

- Booking Completion Rate
- Cancellation Rate
- Total Revenue
- Average Booking Value
- Daily Booking Trend

Tracking these KPIs regularly will enable faster operational decision-making and improve business performance.

CUSTOMER :

1. Improve Customer Satisfaction

The average customer rating is 4.0/5, indicating generally positive service quality. However, there is room to improve by focusing on reducing waiting time and minimizing ride cancellations to increase customer satisfaction.

2. Reduce Customer Waiting Time

Although the average waiting time is approximately 2.85 minutes, reducing it further through better driver allocation and demand forecasting can enhance the customer experience and encourage repeat bookings.

3. Minimize Customer Cancellations

Customer cancellations should be analyzed by cancellation reason and location. Addressing recurring issues such as long waiting times, driver unavailability, or fare-related concerns can improve ride completion rates.

4. Improve Payment Data Quality

A significant portion of bookings appears to have missing payment method information. Improving payment data capture will enable better customer segmentation and support more informed marketing and operational decisions.

5. Reward High-Value Customers

Use the Top Customers analysis to introduce loyalty programs, personalized offers, or priority support for frequent riders to increase customer retention and lifetime value.

6. Monitor Customer Ratings

Analyze low-rated rides to identify common service issues such as driver behavior, delays, or ride quality. Regular monitoring and targeted interventions can improve overall service quality.

7. Personalize Customer Engagement

Segment customers based on booking frequency, payment preferences, and ride behavior to deliver personalized promotions and improve customer retention.

8. Establish Customer Experience KPIs

Management should continuously monitor :

- Average Customer Rating
- Average Waiting Time
- Customer Cancellation Rate
- Preferred Payment Methods
- Repeat Customer Activity

Tracking these KPIs will support ongoing improvements in customer satisfaction and loyalty.

PERFORMANCE :

1. Reduce Driver Cancellation Rate

The driver cancellation rate of 17.9% indicates a significant operational challenge. Analyze cancellation patterns by time, location, and vehicle type, and introduce targeted incentive programs to improve driver reliability.

2. Improve Fleet Completion Rate

With a fleet completion rate of approximately 62%, increasing successful ride completions should be a key operational objective. Better ride allocation and driver availability can improve this metric.

3. Optimize Fleet Allocation

Vehicle types generating higher revenue should receive priority during peak demand periods, while lower-performing categories should be evaluated for operational optimization.

4. Maintain Driver Service Quality

The average driver rating of 4.0 indicates generally good customer satisfaction. Continue monitoring low-rated drivers and provide coaching or additional training where necessary.

5. Improve Driver Performance Monitoring

Implement driver performance scorecards that combine :

- Rating
- Cancellation Rate
- Ride Completion Rate
- Customer Feedback

This will help identify high-performing drivers and those needing support.

6. Optimize High-Distance Trips

Analyze long-distance rides to ensure pricing, driver availability, and profitability remain balanced, particularly during peak demand periods.

7. Focus on High-Revenue Vehicle Segments

Allocate additional drivers and promotional efforts toward vehicle categories contributing the highest revenue while monitoring demand across all fleet types.

OPERATIONS :

1. Increase driver supply between 11 AM–1 PM,

As 12 PM is the busiest booking hour. This can reduce rider waiting time and cancellations.

2. Deploy more drivers on Monday, Tuesday, and Wednesday,

Since these are the highest-demand days. Dynamic scheduling can improve booking completion.

3. Prioritize driver allocation on high-demand routes

Such as Majestic–Hennur and Ramamurthy Nagar–Frazer Town to reduce pickup delays.

4. Maintain current turnaround times

(2.8 min vehicle TAT and 1.4 min customer TAT) by continuously monitoring operational efficiency and investigating any sudden increases.

5. Introduce demand-based driver incentives

During peak hours and on high-demand routes to ensure adequate driver availability.

6. Use demand forecasting

To pre-position drivers before the noon rush instead of reacting after demand spikes.

7. Monitor route-level performance weekly.

Focusing on routes with increasing demand to optimize fleet distribution.

8. Evaluate off-peak utilization and consider promotional offers.

During low-demand hours to balance booking volume throughout the day.

REVENUE :

1. Reduce Revenue Leakage

Revenue leakage is approximately **₹15.95M**, indicating a significant amount of potential revenue lost due to ride cancellations and driver unavailability. Prioritize reducing cancellations to improve revenue realization.

2. Focus on High-Revenue Booking Hours

Analyze and strengthen operations during the booking hours generating the highest revenue by deploying additional drivers and maintaining sufficient supply during these periods.

3. Reduce Driver Not Found Cases

A considerable portion of revenue loss appears to be caused by Driver Not Found cases. Improve driver coverage in high-demand areas through dynamic driver allocation and incentive programs.

4. Improve Customer Retention

Implement targeted customer communication and promotional offers to reduce customer-initiated cancellations, helping recover lost revenue.

5. Optimize Dynamic Pricing

Apply dynamic pricing during peak-demand hours to maximize revenue while maintaining a balance between rider demand and driver availability.

6. Monitor Revenue Trends Daily

Track daily revenue fluctuations and investigate sudden drops to identify operational issues, demand changes, or external factors affecting performance.

7. Increase Net Revenue

While total revenue is **₹35.08M**, net revenue is only **₹19.13M** after accounting for revenue leakage. Reducing cancellations and improving ride completion should be a key strategic objective.

8. Create Route- and Hour-Based Revenue Strategies

Identify combinations of booking hours and locations generating the highest revenue and allocate more drivers to these profitable segments.

9. Track Revenue KPIs Continuously

Monitor Total Revenue, Revenue Lost, Net Revenue, and Average Booking Value regularly to evaluate the effectiveness of revenue optimization initiatives.

10. Set a Revenue Recovery Target

Establish quarterly goals to reduce revenue leakage by **10–15%**, which would directly improve net revenue without requiring additional customer acquisition.

Conclusion :

The analysis indicates that inDrive maintains healthy operational turnaround times and consistent booking demand. However, cancellation rates and revenue leakage continue to impact overall business performance, reducing potential net revenue. By improving driver availability during peak periods, minimizing cancellations, optimizing fleet allocation, and strengthening customer retention strategies, the company can significantly improve ride completion rates and profitability.

Overall, this dashboard provides management with actionable insights to support operational planning, revenue optimization, and customer experience improvements. The recommended actions can help increase efficiency, recover lost revenue, and support sustainable business growth through informed, data-driven decisions.

12. Expected Business Impact

The implementation of the recommended strategies is expected to deliver the following business outcomes:

- **Increase ride completion rate :**
By improving driver availability during peak hours and high-demand routes.
- **Reduce cancellation rates :**
Through targeted customer communication and driver incentive programs.
- **Recover revenue leakage :**
By minimizing cancellations and improving ride fulfillment, resulting in higher net revenue.

- **Improve customer satisfaction :**
By reducing waiting times and maintaining healthy turnaround times.
- **Enhance driver productivity :**
Through optimized fleet allocation and demand-based scheduling.
- **Strengthen operational efficiency :**
By leveraging booking trends and demand forecasting for better resource planning.
- **Support data-driven decision-making :**
By providing leadership with real-time KPIs and performance insights.
- **Increase overall profitability**
Through higher ride completion, improved customer retention, and better revenue optimization.
- **Improve resource utilization**
By allocating drivers and vehicles based on demand patterns rather than static schedules.
- **Enable sustainable business :**
Growth by continuously monitoring performance metrics and acting on analytical insights.

13. Project Deliverables

- **Excel Workbook:** Cleaned dataset, validation checks, pivot tables, exploratory analysis.
- **SQL Script:** Well-organized queries grouped into business modules with comments and reusable views/CTEs.
- **Power BI Report:** Five interactive dashboards with drill-throughs, slicers, KPI cards, and DAX measures.
- **Business Report:** Executive summary, key findings, recommendations, and business impact.

Dataset Disclaimer

Note:

This project was developed for learning and portfolio purposes only. The dataset used in this analysis was obtained from a publicly available YouTube tutorial/source and does not represent official inDrive data.

The project name "inDrive Analytics" has been used solely for demonstration purposes to showcase SQL, Power BI, and data analysis skills. All insights are based on the sample dataset and should not be interpreted as actual business performance of inDrive.

Dataset Source: YouTube tutorial

(<https://youtu.be/1uPUyT9LoHQ?si=fVNstSXtBaKmnSk8>).

DATASET LINK :

(<https://topvarsity.in/ola-data-analyst-project/>)